BOARD BOOK NAVIGATION · COMPARISON RESULTS

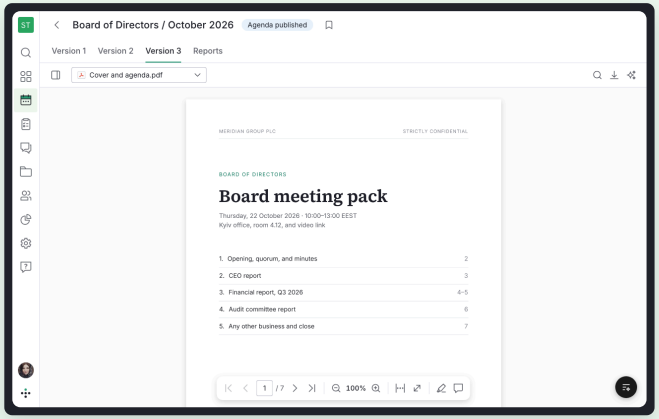
Version 3 wins the comparison

Reading mode is the recommended direction for board book navigation.

Recommended Version 3 gives the best overall experience and leads or ties on all of 5 criteria.

1 Version 3

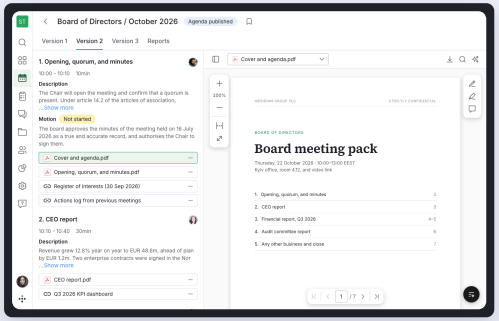
Reading mode



84 / 100

2 Version 2

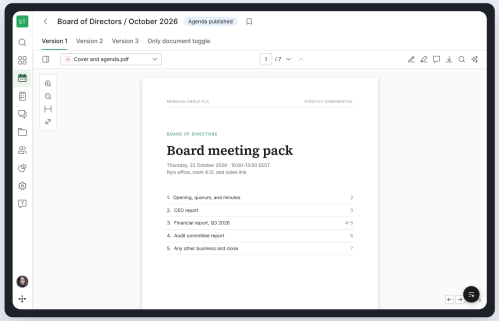
Floating controls



82 / 100

3 Version 1

Full width



65 / 100

2 | **1** | 3

Score breakdown

V3 V2 V1

Criteria	V3	V2	V1
Design review of 30	**27**	25	21
Document boundaries of 5	**4**	4	2
Ease of use (SEQ) of 7	**5.2**	5.1	4.9
Task success %	**94%**	92%	79%
UX score (UXS) of 100	**84**	80	64
Overall mean, 0–100	**84**	82	65

Why Version 3 wins

Next / previous sits exactly where a document ends

Page and document buttons (|< ‹ › >|) live in one bar under the page — no change of focus.

One control zone

Every reading action in one always-visible bar keeps cognitive load low.

Best or tied on all of 5 criteria

Highest overall score (84) with 94% task success.

Next

Version 2 (former Version 4) was re-scored on 2026-10-22 and Version 3, without its progress bar, on 2026-10-23 — both as built; Version 1 reflects its state before the 2026-10-19 fixes. Agenda open by default and the phone "Document tools" sheet count for every version (product decision 2026-10-23). All scores are simulated — validate with directors.

UX AUDIT · BOARD PORTAL

Board Book Navigation — UX Audit

Board UX audit (manual) · 5 lenses · Board personas (Margaret, David, Patricia, Robert) · grounded in Sauro's Big 4, Laws of UX, Shneiderman's 8 Golden Rules, Nielsen's 5 quality components · 2026-10-23

> There is no board-specific ux-audit skill in this session; the FVDR ux-audit (Chris, Jim… and VDR data) does not apply to Board. This audit follows the same 5-lens structure with Board personas. Scores are expert judgement on the prototype plus the simulated task success from the usability study — not measured analytics. No Clarity or NPS data was available. Re-run 2026-10-22: Version 2 was replaced by the former Version 4 and re-scored; V1 / V3 scores are unchanged. Re-run 2026-10-23: Version 3 lost its reading-progress bar and was re-scored as built; both V2 and V3 now reflect their current state. Product decision 2026-10-23: agenda open by default and one phone tools sheet in every version — neither counts as a V2-only strength.

Intent. Readers must (1) know which document and page they are on, (2) read a document continuously, (3) move to the next document — possible only through the viewer's navigation panels because continuous scroll stops at each document's end — and (4) jump to an agenda item's documents, annotate and search.

01 UX Score (UXS)

V1 Full width

64 / 100 UXS

Not recommended

Familiar for power users, but the top-only navigation fails low-proficiency and touch readers at document boundaries.

V2 Floating controls

80 / 100 UXS

Close second

Same bottom-bar navigation as V3; with the agenda and the phone sheet now shared by every version, what remains is three control zones on desktop versus V3's one.

V3 Reading mode

84 / 100 UXS

Recommended — validate

One control zone and a quiet top bar — the lowest cognitive load, with the agenda open by default and the shared phone sheet. Validate with R1

(labelled 'Next document') and R3 (local + global position).

UXS = mean of the 5 lens scores (1–5) × 20. 1 = poor · 3 = acceptable with friction · 5 = excellent.

02 Lens scores

Lens	V1 Full width	V2 Floating controls	V3 Reading mode
Intent alignment Does the layout serve the reader's real job under the one-document-at-a-time constraint?	**3 / 5** Optimised for a PDF-editor mental model (top toolbar); the 'continue' action sits far from where reading ends.	**4 / 5** Same reading flow as V3 (one bottom pill for page / document), with zoom and annotation tools split into two more floating cards.	**4 / 5** Built around reading: document first, one bar for every reading action (page, document, zoom, annotations).
Findability Can each persona find the next-document action, the agenda and the page jump without help?	**3 / 5** Agenda behind an unlabeled icon; ⌄ / ^ read as dropdown/collapse (Jakob's law — page turns are ‹ ›).	**4 / 5** Agenda open by default; ‹ › and |< >| in one predictable pill — but controls spread over three zones and all are icon-only.	**4 / 5** ‹ › |< >| and the page field in one predictable bar; agenda open by default (product decision). Icon-only controls.
Cognitive load How many places must the reader scan, and how much must they remember?	**3 / 5** Two control zones (top toolbar + left card); icon-only tools.	**3 / 5** Three control zones on desktop (view card, palette, bottom pill) instead of four; one button on phones. Still more to scan than V3's single bar.	**4 / 5** One bottom bar + quiet top bar; Shneiderman 'reduce short-term memory load' is met best.
Interaction quality Are controls reachable, consistent and giving feedback (desktop, iPad, phone)?	**3 / 5** Fitts's law: the next control is at the top while the pointer/eye is at the bottom; mobile tools in ⋯.	**4 / 5** Pill in the thumb zone; 'Fit' shown when fitted; pressed controls keep the hover fill. Tablet portrait still	**4 / 5** Bottom bar in the thumb zone on iPad; consistent ‹ ›; no feedback that 'next' changes document.

Lens	V1 Full width	V2 Floating controls	V3 Reading mode
		shows the two cards; pill buttons are 28px.	
Outcome quality Do readers complete the tasks? (Sauro's Big 4: completion, time, errors, satisfaction — from the simulated study)	**4 / 5** Simulated success 77%, SEQ 4.5 / 7; Patricia fails US03 (thinks the pack ended).	**5 / 5** Simulated success 92%, SEQ 5.1 / 7; no failures.	**5 / 5** Simulated success 92%, SEQ 4.9 / 7; no failures.
UXS	**64 / 100**	**80 / 100**	**84 / 100**

03 Ship-readiness risks

Document end stays silent — the bar must say 'Next document: <name>' on a document's last page (study R1). V2 and V3 have >| 'Next document: <name>' (tooltip only, icon button).

No variant shows how much is left since the progress bar was removed; global 'N / 7' without a local page count misleads about document length (study R3).

Icon-only controls exclude low-proficiency and touch users (study R4) — incl. V2's 6-dot phone trigger and V3's ⋯.

Agenda open by default and the shared phone sheet are product decisions (2026-10-23) not yet built into the V1 / V3 prototypes — implement before testing with real directors.

USABILITY EVALUATION REPORT · BOARD PORTAL

Board Book Navigation — Usability Evaluation Report

Synthetic Persona Usability Study | NN/G Methodology | 4 Personas | 10 User Stories | 3 Variants · 2026-10-23

> **Synthetic study.** All results come from simulated think-aloud sessions with synthetic personas. Numbers are simulated, not measured user data, and must be validated with real users.

Prototype at localhost:5173/board-book-page-nav.html with the real Q4 board pack (5 agenda items, 6 documents, 7 pages). Platform constraint: continuous scroll works only within one document; nothing can be added inside the document, so moving to the next document is possible only through the viewer's navigation panels ('next page' on the last page, the document dropdown, keyboard ← →). State evaluated: after the 4 Must-fix items from the design review. Re-run 2026-10-22: Version 2 was replaced by the former Version 4 (bottom page pill with previous / next document, zoom card left, annotation palette right, agenda open, phone 'Document tools' sheet); its column was re-simulated with the same personas and stories. Since 2026-10-19 all versions also lost the '← → pages' keyboard hint (the keys still work) and got the Figma Search / Ask AI panels; V1 / V3 ratings were not re-simulated. Re-run 2026-10-23: the reading-progress bar was removed from Version 3, and Version 3 was re-rated in its current state (|< / >| document buttons in the bottom bar, no progress) with the same rules as the new Version 2, so both are compared as built. Product decision 2026-10-23: in every version the agenda is open by default and the phone tools open from the same 'Document tools' (dots) sheet. US04 and the phone story were re-rated on that basis (V1 / V3 prototypes do not implement it yet).

01 Executive summary

V1 Full width	V2 Floating controls	V3 Reading mode · best	Issues
4.9 / 7 SEQ	**5.1** / 7 SEQ	**5.2** / 7 SEQ	**1** Critical
Task success 79%	Task success 92%	Task success 94%	4 High · 2 heuristics violated

CRITICAL Document end is silent and the only way on is a control labelled 'Next page' — Patricia stops reading in V1; David hesitates.

HIGH V1 places next/previous in the top toolbar with vertical ⌄ / ^ icons.

HIGH Global counter 'N / 7' beside a single-document view.

STRENGTH Version 3 keeps every navigation action in one always-visible bar where reading ends — highest task success (94%) and best SEQ for Patricia and Robert.

STRENGTH With the agenda open by default in every version (product decision 2026-10-23), jumping to an agenda item's documents is effortless for all personas (US04).

02 Personas

MC Margaret Chen
Corporate Secretary · High

SEQ V1 / V2 / V3

6.3 / 6.0 / 5.8

Success V1 / V2 / V3

100% / 100% / 100%

Goal

Check the final board pack end to end before it goes to directors.

Mental model

Keyboard shortcuts, dense toolbars, Notion/Acrobat-like patterns.

Risk behaviour

Impatient with extra steps; skims past labels.

Emotional profile

DO David Okafor
Corporate Secretary · Medium

SEQ V1 / V2 / V3

4.7 / 5.0 / 5.0

Success V1 / V2 / V3

83% / 92% / 92%

Goal

Review the pack and annotate items that need the Chair's attention.

Mental model

Explicit toolbar buttons and labelled Next / Previous.

Risk behaviour

Hesitates when something changes without explanation.

Emotional profile

Cautious; fears missing a document in a legal pack.

PR Patricia Reyes
Corporate Secretary · Low

SEQ V1 / V2 / V3

3.8 / 4.7 / 4.8

Success V1 / V2 / V3

50% / 75% / 83%

Goal

Open the pack and read the documents for one agenda item.

Mental model

A visible, labelled button for every action; step by step.

Risk behaviour

Ignores icon-only controls; never opens ⋯ menus.

Emotional profile

Anxious; needs plain language.

RH Robert Hale
Director / Board Member · Occasional

SEQ V1 / V2 / V3

4.7 / 4.8 / 5.0

Success V1 / V2 / V3

83% / 100% / 100%

Goal

Read the pack on iPad the evening before the meeting and mark questions.

Mental model

Reading app: swipe or tap to turn pages, everything within thumb reach.

Risk behaviour

No keyboard; low patience for admin-style UI.

Emotional profile

Methodical; low tolerance for ambiguity.			Time-poor; wants the right document fast.

03 Variants, research goals and user stories

V1 Full width: Top toolbar: TOC toggle · document dropdown · page input N / 7 with ⌄ next / ^ previous · tools. Vertical zoom card on the left. Agenda closed by default. Below 768px tools collapse into ⋯.

V2 Floating controls: Agenda open beside the document (collapses while Search / Ask AI is open below 1800px). Top bar: agenda toggle · document dropdown · download / search / Ask AI. One floating view card left (+ % − | fit width · fullscreen, icon buttons with tooltips), annotation palette right (draw, highlight, comment). One bottom pill: |< previous document · ‹ [N] / 7 › · next document >| with 'Previous / Next document: <name>' tooltips. Phone (<768px): the cards fold into one 'Document tools' button (6 dots) opening a labelled DS bottom sheet (Zoom − % +, Fit to width, Fit to page, Actual size, Fullscreen, Annotate, Add comment). Re-simulated 2026-10-22.

V3 Reading mode: Top bar: TOC toggle · document dropdown · search / download / Ask AI. Always-visible bottom bar: |< previous document · ‹ N / 7 › · next document >| · zoom − % + · fit width · focus mode · highlight / comment (zoom, fit, focus and annotation tools in ⋯ below 768px). Agenda closed by default. No reading-progress bar (removed 2026-10-23). Re-rated 2026-10-23 in this state.

G1 Assess orientation: can each persona tell which document and page they are on (H1 visibility of system status).

G2 Identify breakdown moments at document boundaries under the one-document-at-a-time constraint.

G3 Evaluate control placement: travel distance from the point of reading to the 'continue' action, per variant.

G4 Measure cognitive load of each layout (number of control zones, icon-only actions, hidden affordances).

G5 Quantify simulated task success and SEQ per user story, persona and variant.

US01 As a reader, I want to see which document and page I am on, so I don't lose my place. (G1)

US02 As a reader, I want to read a document page by page without hunting for controls. (G3, G4)

US03 As a reader, I want to continue to the next document when I reach the end of the current one. (G2, G3)

US04 As a reader, I want to jump straight to the documents of a specific agenda item. (G4)

US05 As a reader, I want to go to a page number someone mentions in the meeting. (G1, G5)

US06 As a reader, I want to zoom or fit a dense table so I can read the figures. (G4)

US07 As a reader, I want to highlight or comment on a passage to discuss it later. (G4)

US08 As a reader, I want to search the board book for a term. (G4)

US09 As a reader, I want to read comfortably on a tablet or phone. (G3, G4)

US10 As a reader, I want to go back to the previous document. (G2)

04 Task simulation

Margaret Chen — Corporate Secretary, High

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US01	Opens the pack to spot-check the cover.	Reads the dropdown and N / 7 instantly in all variants.	V2/V3: document name (top bar) and page (bottom bar) sit far apart.	6 / 5 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US03	Finishes 'Opening, quorum, and minutes' and wants the CEO report.	Presses → on the keyboard; the next document loads.	None for her — the keyboard hint tells her ← → works. No friction: → on the keyboard (the on-screen hint is gone, she knows the keys) or >| in the pill.	6 / 6 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US04	Jumps to item 4 to verify the audit motion wording.	Agenda is open by default in every version: clicks the document in the agenda list.	None — agenda open by default (product decision 2026-10-23).	7 / 7 / 7	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US05	'Look at page 6' — types the number.	Types 6 in the page field.	V2/V3: the page field is in the bottom bar; she looks at the top first.	7 / 6 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US06	Checks the cash-flow table on page 5.	Uses zoom-in; then Fit width.	V2: the card now shows 'Fit' when fitted, so the old '100%' confusion is gone.	6 / 6 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US07	Highlights the auditor recommendation.	Picks Highlight; V3 she also looks for Draw.	V3: no Draw tool.	6 / 6 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success

David Okafor — Corporate Secretary, Medium

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US01	Opens the pack on the cover.	Reads 'Cover and agenda.pdf' and '1 / 7' and assumes the cover has 7 pages.	Global counter next to a single-document view (all variants); no variant shows progress any more.	5 / 4 / 4	V1 ✔ Success V2 ⚠ Partial V3 ⚠ Partial
US03	Reaches the bottom of 'Opening, quorum, and minutes'.	Scrolls, nothing more appears; thinks the pack failed to load. V1: finds ⌄ at the top, unsure it means 'next document'. V2: clicks › beside the sheet. V3: › is right under where he stopped.	Silent document end; 'Next page' tooltip never says a new document opens. V2: the bottom pill is right under where he stopped; >| names the next document on hover. Still nothing in the page itself.	3 / 5 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success
US05	Chair says 'page 6'.	Types 6.	None.	6 / 5 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US06	Zooms the KPI table.	Clicks + twice.	V2: zoom and view are still two cards, but 'Fit' / '%' make the state clear.	5 / 5 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US07	Comments on the retention bonus pool.	Selects Comment and clicks the sheet.	No confirmation that the comment pin was saved (all variants).	5 / 6 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US10	Wants to re-check the previous document.	V1: ^ at the top; V2/V3: ‹.	V1: ^ reads as 'collapse', not 'previous document'. V2: |< 'Previous document' sits next to ‹ in the pill.	4 / 5 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success

Patricia Reyes — Corporate Secretary, Low

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US01	Opens the pack for the first time.	Scans for words; finds the document name but not what the number means.	V2: controls in three places (left card, right palette, bottom pill) — fewer than before, still icon-only.	4 / 4 / 5	V1 ⚠ Partial V2 ⚠ Partial V3 ✔ Success
US02	Reads the CEO report page by page.	Looks for a 'next' button. V1: thinks ⌄ opens a list. V2: big round › is obvious. V3: › in the bottom bar.	V1: vertical chevrons look like dropdown toggles. V2: ‹ › in the bottom pill are small (28px) but where she expects them.	3 / 5 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success
US03	Gets to the end of the minutes	V1: believes the pack has ended and stops. V2/V3:	Nothing signals that more documents follow; V1's continue control is far away at the top.	2 / 4 / 4	V1 ✕ Failure V2 ⚠ Partial

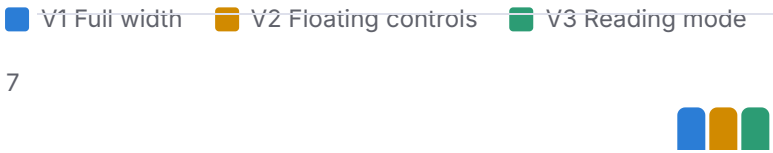
Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
	document.	notices the › arrow after a pause and tries it.	V2/V3: the continue control is in the right place, but |< / >| are icon-only and their names need hover.		V3 ⚠ Partial
US04	Needs the Audit committee report for item 4.	Agenda is open by default in every version: clicks the document in the agenda list.	None — agenda open by default (product decision 2026-10-23).	6 / 6 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US06	The financial table is too small.	Looks for a magnifier with '+'.	V1: icon-only card; V3: '– 100% +' reads clearly. V2: '+ 100% –' in the left card reads clearly, as in V3.	4 / 5 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success
US09	Opens the pack on her phone.	Every version opens its tools from the same 'Document tools' (dots) sheet; rows are labelled once open.	The dots trigger is icon-only, so she hesitates to open it (all versions).	4 / 4 / 4	V1 ⚠ Partial V2 ⚠ Partial V3 ⚠ Partial

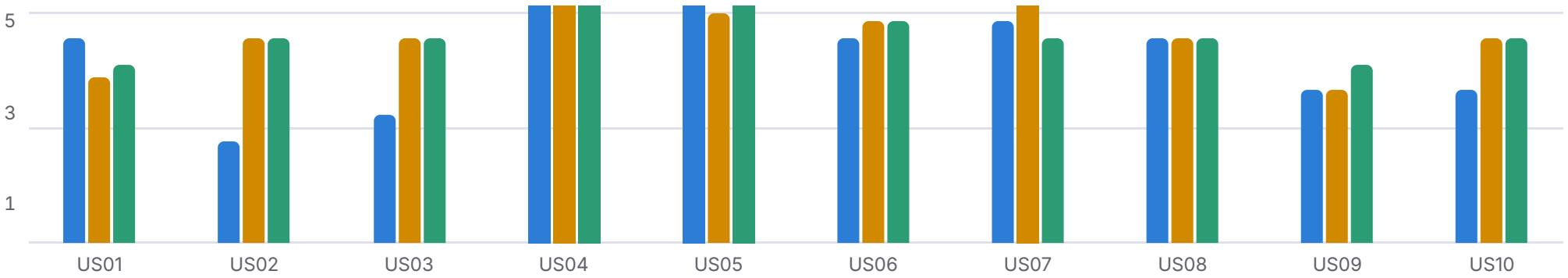
Robert Hale — Director / Board Member, Occasional

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US01	Opens the pack on iPad the night before.	Wants to know how much is left.	No variant shows how much is left; V2/V3 show the counter in the bottom bar.	5 / 4 / 4	V1 ✔ Success V2 ✔ Success V3 ✔ Success

Task	Scenario	Behaviour observed	Friction	SEQ V1 / V2 / V3	Outcome V1 / V2 / V3
US03	Finishes the CEO report, wants the financials.	No keyboard. V1: has to reach the top toolbar. V2: taps › beside the sheet. V3: taps › in the bottom bar (thumb zone).	V1: continue control outside thumb reach and unlabeled. V2/V3: › and >| in the bottom pill, in the thumb zone.	3 / 5 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success
US04	Goes straight to the audit committee report.	Agenda is open by default in every version: clicks the document in the agenda list.	None — agenda open by default (product decision 2026-10-23).	6 / 6 / 6	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US07	Marks a question on the auditor fee.	Highlight, then comment.	Icon-only tools; tooltips don't show on touch.	5 / 5 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US08	Searches for 'Halden'.	Taps the search icon.	None significant.	5 / 5 / 5	V1 ✔ Success V2 ✔ Success V3 ✔ Success
US09	Reads on iPad in portrait.	Scrolls with one hand.	V2: on iPad portrait (768–1023) the two cards stay in the side gutters beside the sheet — no overlap, a little visual noise.	4 / 4 / 5	V1 ⚠ Partial V2 ✔ Success V3 ✔ Success

05 Simulated metrics





Average SEQ (1–7) per user story, simulated. Exact values in the table below.

Story	V1 Full width	V2 Floating controls	V3 Reading mode
US01	5.0	4.3	4.5
US02	3.0	5.0	5.0
US03	3.5	5.0	5.0
US04	6.3	6.3	6.3
US05	6.5	5.5	6.0
US06	5.0	5.3	5.3
US07	5.3	5.7	5.0
US08	5.0	5.0	5.0
US09	4.0	4.0	4.5
US10	4.0	5.0	5.0

Persona	V1 Full width — SEQ · success · time (fast / med / slow)	V2 Floating controls — SEQ · success · time (fast / med / slow)	V3 Reading mode — SEQ · success · time (fast / med / slow)
Margaret Chen	6.3 · 100% · 5 / 1 / 0	6.0 · 100% · 5 / 1 / 0	5.8 · 100% · 4 / 2 / 0
David Okafor	4.7 · 83% · 3 / 2 / 1	5.0 · 92% · 4 / 2 / 0	5.0 · 92% · 4 / 2 / 0
Patricia Reyes	3.8 · 50% · 0 / 2 / 4	4.7 · 75% · 2 / 3 / 1	4.8 · 83% · 2 / 3 / 1
Robert Hale	4.7 · 83% · 3 / 1 / 2	4.8 · 100% · 4 / 2 / 0	5.0 · 100% · 4 / 2 / 0
Overall	**4.9 / 7 · 79%**	**5.1 / 7 · 92%**	**5.2 / 7 · 94%**

06 Cross-persona pattern analysis

Shared breakdown points

The end of a document is silent in every variant: nothing below the last page and the 'next' control is labelled 'Next page', so no persona is told that another document follows.

The global counter 'N / 7' sits next to a single-document view, so David and Patricia misread how long the current document is.

All navigation and tool controls are icon-only (incl. V2's |< >| and its 6-dot 'Document tools' button); tooltips need hover, so they don't help Patricia or Robert on iPad.

Divergent experiences

Interaction	Margaret (High)	David (Medium)	Patricia (Low)	Robert (Occasional)
Continue to next document	Keyboard → — no friction	Hesitates; V3 easiest	V1: gives up; V2/V3: finds › / >| in the bottom bar after a pause	Needs thumb reach — V2/V3 bottom bar

Interaction	Margaret (High)	David (Medium)	Patricia (Low)	Robert (Occasional)
Agenda (contents)	Uses it — open by default	Uses it — open by default	Uses it — open by default	Uses it — open by default
Zoom / fit	Any variant	V2: two cards remain, 'Fit' label helps	Reads V3's '– 100% +'	Pinch-zoom expected
Mobile tools (Document tools sheet)	Opens it	Opens it after a pause	Hesitates at the icon-only dots trigger	Rarely needs it on iPad

07 Heuristic assessment (NN/G)

Heuristic	Status	Evidence
H1 Visibility of system status	VIOLATED	Document end is silent; global N / 7 misread as document length (US01, US03).
H2 Match with the real world	PARTIAL	V1 ⌄ / ^ read as dropdown/collapse, not page turns (US02, US10).
H3 User control and freedom	PASS	Previous page on the first page returns to the previous document; Esc leaves focus mode.
H4 Consistency and standards	PARTIAL	V1 uses ⌄^, V2/V3 ‹ › for pages and |< >| for documents; zoom % still means different things in V3 on phones (US06).
H6 Recognition rather than recall	VIOLATED	Icon-only controls, agenda behind an unlabeled icon, tools in ⋯ on mobile (US04, US09).
H7 Flexibility and efficiency	PASS	Keyboard ← → crosses documents; page field accepts direct input (US03, US05).
H8 Aesthetic and minimalist design	PARTIAL	V2 now uses three zones (view card, palette, bottom pill) and one button on phones; V3 consolidates everything in one bar.

Heuristic	Status	Evidence
H9 Help users recover from errors	PARTIAL	When Patricia believes the pack ended, nothing helps her recover (US03, V1).

08 Severity matrix

ID	Tier	Issue	Freq.	Impact	Persistence	Personas	Heuristics
I-01	Critical	Document end is silent and the only way on is a control labelled 'Next page'	3/4	Patricia stops reading in V1; David hesitates	Stays stuck (PR, V1)	DO, PR, DIR	H1, H9
I-02	High	V1 places next/previous in the top toolbar with vertical ⌄ / ^ icons	3/4	Far from the reading end point; icons misread	Retries	DO, PR, DIR	H2, H4
I-03	High	Global counter 'N / 7' beside a single-document view	2/4	Wrong sense of document length and position	Self-recovers slowly	DO, PR	H1, H2
I-04	High	Icon-only navigation and tools (incl. V2 |< >|); tooltips need hover	2/4	Excludes low-proficiency and touch users	Retries	PR, DIR	H6
I-05	High	Agenda closed by default behind an unlabeled sidebar icon — resolved: open by default in every version (product decision 2026-10-23)	2/4	Patricia can't jump to an agenda item's documents	Retries	PR, DIR	H6
I-06	Medium	V2 spreads controls over three zones; on tablet (768–1023) the cards stay in the gutters beside the sheet	2/4	Scanning cost; some clutter at the sheet edges	Self-recovers	PR, DIR	H8

ID	Tier	Issue	Freq.	Impact	Persistence	Personas	Heuristics
I-07	Medium	Zoom % meaning differs (V3 '54%' default on phone); V2 now shows 'Fit' when fitted	1/4	Unsure what 'actual size' is	Self-recovers	PR	H4
I-08	Medium	Mobile: tools behind an icon-only trigger — now one labelled 'Document tools' sheet in every version, but the dots trigger itself is unlabeled	1/4	Patricia never finds Highlight on the phone	Stays stuck	PR	H6
I-09	Low	V3 focus-mode icon (⌃⌄ 'expand') is ambiguous — fixed 2026-10-20: arrow-resize-diagonal, moved to the bottom bar	1/4	Rarely used	Self-recovers	DO	H2
I-10	Low	V3 has no Draw tool	1/4	Feature-parity gap for power users	Self-recovers	MC	H7

09 Prioritized recommendations

R1 Make the document boundary visible inside the navigation panel

On the last page of a document the next control becomes a labelled DS Button 'Next document: <name> ›' (and 'Previous document' on the first page); tooltip says the same.

Effort: Low · Impact: DO, PR, DIR · **Target:** US03 success 71% → 90%+; PR US03 SEQ 3.3 → 5.0+

Status: V3, 2026-10-20: partly done — ‹‹ / ›› buttons with 'Previous / Next document: <name>' tooltips are in the bottom bar; not re-tested

R2 Use V3's bottom bar as the primary navigation location

Keep ‹ N / 7 › at the bottom where reading ends; in V1 replace ⌄ / ⌃ with ‹ ›.

Effort: Med · Impact: All · **Target:** US02 avg SEQ 4.7 → 5.5+

R3 Show local and global position

R4 Label the primary actions

'Page 1 of 2 · 4 / 7' in the counter; document name next to it.

Effort: Low · Impact: DO, PR · **Target:** US01 avg SEQ 4.8 → 6.0

Text labels on Previous / Next at ≥1024px; persistent labels instead of hover-only tooltips on touch.

Effort: Med · Impact: PR, DIR · **Target:** PR task success 61% → 75%+

R5 Open the agenda by default on wide screens

In V3 open the agenda panel by default at ≥1280px (as V2 does), with a labelled toggle 'Agenda'.

Effort: Med · Impact: PR, DIR · **Target:** US04 PR SEQ 3 → 6
Status: Adopted 2026-10-23: agenda open by default in every version

R6 Align zoom semantics and label the mobile menu

Show 'Fit' when fitted (done in V2); 100% = sheet's natural width; label the mobile tools trigger ('More tools' / 'Tools') — V2's sheet already labels every row.

Effort: Low · Impact: DO, PR · **Target:** US06 avg SEQ 4.9 → 5.5

R7 Carry V2's phone sheet to tablet portrait and to V3

Use the labelled 'Document tools' bottom sheet below 1024px (not only below 768px) and in V3's mobile bar instead of ⋯; give the trigger a text label.

Effort: Low · Impact: PR, DIR · **Target:** US09 PR SEQ 4 → 5+; DIR US09 4 → 5
Status: Adopted 2026-10-23: the dots 'Document tools' sheet is the phone pattern for every version

10 Next testing iteration

Phase 1 — Validation: After sprint 1, re-test I-01…I-05 fixes on V3 and the new V2 with the same persona types.

Phase 2 — Breadth: Add iPad and phone tasks; recruit 2 real users per persona type (incl. 2 directors).

Phase 3 — Longitudinal: Learnability over 3 meeting cycles; diary study with directors reading packs at home.

Open research questions

Do directors read packs linearly or jump by agenda item?

Is a labelled 'Next document' enough, or do readers expect the next document to load automatically?

How often do readers need the previous document mid-meeting?

Which share of readers uses the keyboard at all?

Do readers miss a 'how much is left' signal now that no variant shows reading progress?

---

11 Takeaways

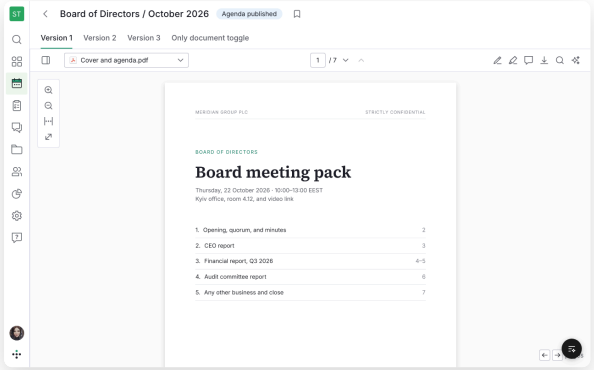
With the agenda open by default and the same phone sheet in every version, Version 3 leads again: its single control zone is the difference that remains.

Version 2 is a close second — same bottom-bar navigation, but three control zones on desktop.

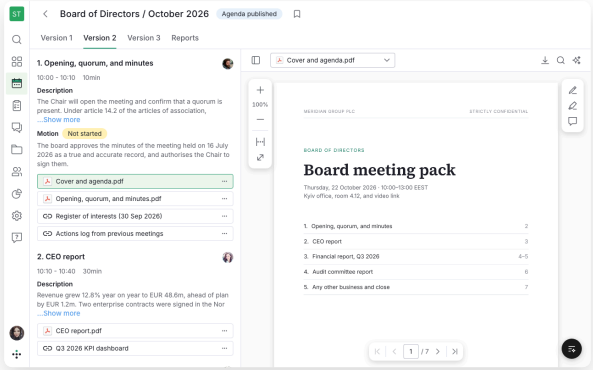
The document boundary is the critical moment — the fix belongs in the navigation panel, not the document.

Labels and position (local + global) matter more than more controls.

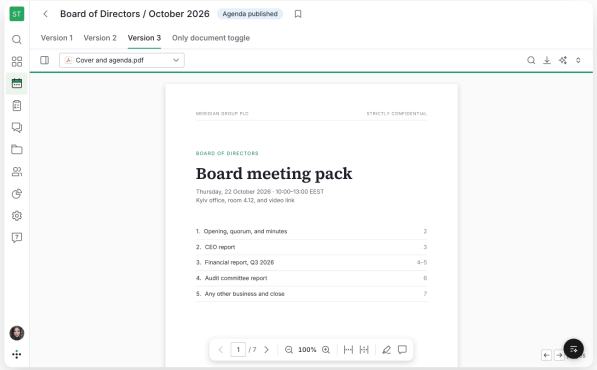
Validate with real directors on iPad before committing — the gap is small and the study is simulated.



Version 1 — Full width



Version 2 — Floating controls



Version 3 — Reading mode

Design Review: Board book — document navigation (Version 1, 2, 3)

Reviewed against: there is no dedicated `DESIGN_BRIEF.md` for this feature. The reference is the task brief from chat (the Version 1 / 2 / 3 descriptions), the Figma mockups (Meetings › 32114-124836, "Navigation" 32133-81902) and the DS rules in `CLAUDE.md`.

Philosophy: Board DS (Ideals) — a calm, utilitarian interface with the document at the centre.

Date: 2026-10-19 · Version 2 re-reviewed 2026-10-22 · Version 3 re-reviewed 2026-10-23

Page: `localhost:5173/board-book-page-nav.html?tab=v1|v2|v3`, document — Q4 board pack (7 pages).

> This review covers DS fidelity, layout and heuristics. The persona-based evaluation (board-usability-study method) is in the **Usability study** section of this page, and the 5-lens scoring is in **UX audit**.

01 Screenshots Captured

Screenshot	Breakpoint	Description		
`screenshots/review-v1-desktop-1280.png`	Desktop (1280×800)	V1 — full width, toolbar with page counter, zoom card on the left		
`screenshots/review-v2-desktop-1280.png`	Desktop (1280×800)	V2 (new, 2026-10-22) — agenda open; one view card left, palette right, bottom pill \	< ‹ 1 / 7 › >\	
`screenshots/review-v3-desktop-1280.png`	Desktop (1280×800)	V3 (before 2026-10-20) — reading mode, bottom bar, progress bar		
`screenshots/review-v1-tablet-768.png`	Tablet (768×1024)	V1 — toolbar fits, agenda closed		
`screenshots/review-v2-tablet-768.png`	Tablet (768×1024)	V2 (new) — agenda closed; cards in the gutters beside the sheet, pill at the bottom		
`screenshots/review-v3-tablet-768.png`	Tablet (768×1024)	V3 — bar fits; progress shown as 100% (see Should fix 1)		
`screenshots/review-v1-mobile-375.png`	Mobile (375×812)	V1 — document field collides with the page field; the 635px sheet doesn't fit		
`screenshots/review-v2-mobile-375.png`	Mobile (375×812)	V2 (new) — one "Document tools" button; sheet fits; pill fits		
`screenshots/review-v2-mobile-sheet-375.png`	Mobile (375×812)	V2 (new) — "Document tools" bottom sheet with labelled rows		
`screenshots/review-v3-mobile-375.png`	Mobile (375×812)	V3 — sheet fits, but the bottom bar is cut off on the right		
`screenshots/fixed-v1-mobile-375.png`	Mobile (375×812)	After the fix: V1 — document · 1 / 7 · ⌄ ^ · "⋯" all fit		

Screenshot	Breakpoint	Description
`screenshots/fixed-v2-desktop-1280.png`	Desktop (1280×800)	Old V2 (removed 2026-10-22) — after the fix: controls visible without hover, "1 / 7" counter at the bottom
`screenshots/fixed-v2-tablet-768.png`	Tablet (768×1024)	Old V2 (removed) — same on tablet
`screenshots/fixed-v3-mobile-375.png`	Mobile (375×812)	After the fix: V3 — ‹ 1 / 7 › · zoom · "⋯" in one bar
`screenshots/updated-v3-desktop-1280.png`	Desktop (1280×800)	Update 2026-10-20: V3 — ‹‹ ‹ 1 / 7 › ›› · zoom · Fit width · Focus · annotations in the bottom bar; top bar has only Search / Download / Ask AI
`screenshots/updated-v3-tablet-768.png`	Tablet (768×1024)	Update 2026-10-20: V3 — same bar on tablet

> Screenshots are in `.design/board-book-page-nav/screenshots/`. Desktop and tablet — headless Chrome; mobile — the in-app browser (headless Chrome distorts the 375 width; the 2026-10-22 mobile captures were rescaled from the browser pane). Dark mode was not checked: the test page has no theme toggle.

02 Summary

> **Product decision 2026-10-23:** in every version the agenda is **open by default** and the phone tools open from the same **"Document tools" (dots) sheet**. Both are scored as shared, not as V2 strengths (the V1 / V3 prototypes don't implement them yet). **Update 2026-10-23 — Version 3 re-reviewed.** The reading-progress bar was removed from Version 3. Its column below is now re-scored in its current state (bottom bar with \|< ‹ N / 7 › >\|, zoom, fit width, focus, annotations; no progress), the same way as the new Version 2, so V2 and V3 are compared as built. See `screenshots/updated-v3-desktop-1280.png`.

> **Update 2026-10-22 — Version 2 replaced.** The original "floating controls" Version 2 (round ‹ › arrows beside the sheet, counter at the bottom) was removed. **Version 2 is now the former Version 4:** agenda open beside the document; one view card on the left (+ % − | Fit width · Fullscreen icon buttons with tooltips, merged 2026-10-22); the annotation palette on the right (draw · highlight ·

> comment); **one bottom pill** `|< ‹ [N] / 7 › >|` with "Previous / Next document: <name>" tooltips; on phones (<768px) both cards fold into one **"Document tools"** button that opens a labelled DS bottom sheet (Zoom − % +, Fit to width, Fit to page, Actual size, Fullscreen, Annotate, Add comment). Pressed controls keep the hover fill. The V2 column below was re-scored; V1 / V3 scores are unchanged. Shared changes since 2026-10-19 in all versions: the "← → pages" keyboard hint was removed (keys still work), Search / Ask AI panels follow Figma, and the agenda collapses while Search / Ask AI is open below 1800px.

All three versions are built from DS components, and continuous scrolling with the real document works in all of them. **Version 3** is the strongest base: every key action (page, zoom, annotations) lives in one always-visible place, there is reading progress, and on mobile the sheet fits. **Version 2 (new)** is a close second: it has the same bottom-bar navigation as V3, but three control zones instead of one. Its open agenda and phone sheet are now the default for every version. Since V3 lost its progress bar, neither shows how much is left. **Version 1** is the most familiar PDF pattern, but its toolbar breaks on mobile.

> **Update 2026-10-19:** all 4 Must-fix items have been fixed (status in each item below). Scores in the table reflect the state before the fixes.

> **Update 2026-10-20 — V3 bottom bar:** (1) **Previous / Next document** buttons (‹‹ ››, DS `angle-double-left/right-fill`) were added around the page controls; their tooltip names the target document ("Next document: CEO report.pdf") and they are disabled on the first / last document. (2) "Fit width" and "Fit page" were merged back into one **Fit width** button. (3) **Focus mode** moved from the top bar to the bottom bar, with the `arrow-resize-diagonal` icon. The top bar now holds only Search, Download and Ask AI. Below 768px zoom, Fit width, Focus mode and annotations live in the "⋯" menu. See `screenshots/updated-v3-desktop-1280.png`. Scores were not re-rated; the change closes Should fix 5 and most of the document-boundary point 4.

> **Update 2026-10-19 — platform constraint:** continuous scrolling works only **within one document**. Nothing can be added inside the document (for example, a "Next document" button under the last page). Moving to the next document is possible **only through the viewer's navigation panels**: "next page" on the last page, the document dropdown, the keyboard. The prototype now follows this behaviour in every version: `PageStack` renders only the current document's pages. This makes **the placement of the navigation panels the key criterion**. Details in "Document boundaries" below.

03 Comparison

Scores 1–5 (5 = best).

Criterion	V1 Full width	V2 Floating controls (new)	V3 Reading mode				
Where am I (page / document)	5 — "1 / 7" + document in the toolbar	4 — document in the top bar, "1 / 7" in the bottom pill; no progress	4 — document in the top bar, "1 / 7" in the bottom bar; no progress				
Findability of controls	4 — everything visible, but two zones (toolbar + card)	4 — always visible, three zones (view card, palette, pill)	5 — one persistent bar				
Agenda context	5 — open by default (product decision)	5 — open next to the document	5 — open by default (product decision)				
Reading space	4	4 — cards sit in the gutters beside the sheet; one button on phones	5				
Mobile (375)	1 — toolbar collides, sheet doesn't fit	4 — sheet and pill fit; labelled "Document tools" sheet	4 — sheet and bar fit; tools in an unlabeled ⋯				
Consistent "next / back" semantics	2 — ⌄ / ^ while the hint says ← →	4 — ‹ › for pages, \	< >\	for documents	4 — ‹ › for pages, \	< >\	for documents
Total	**21**	**25**	**27**				

> V1 scores reflect the state before the 2026-10-19 fixes; V2 (2026-10-22) and V3 (2026-10-23) are scored as built, with the same rules.

Recommendation: V3 is the base — one control zone and a quiet top bar, with the agenda open by default and the shared "Document tools" sheet on phones (product decision 2026-10-23). V2 is a close second; its extra floating cards are the main difference.

04 Document boundaries (after the constraint was clarified)

A document ends "silently": there is nothing under the last page, and the only way on is a control in a navigation panel. So what matters is **where that panel sits relative to the point where the reader finishes the document**: the bottom of the last page.

What matters at a document boundary	V1 Full width	V2 Floating controls (new)	V3 Reading mode		
Where "next page / document" is when you reach the bottom	at the top, in the toolbar (⌄) — eyes and pointer must travel across the whole screen	at the bottom, in the pill (› / >\	) — exactly where you finished reading	at the bottom, in the bar (›) — exactly where you finished reading	
Control visibility	always	always	always		
How you can tell the document ended	page counter / name in the document field only	page counter / document name (no progress)	page counter / document name (no progress)		
Previous document	^ on the first page (top)	‹ on the first page, or \	< from any page (bottom)	‹ on the first page, or \	< from any page (bottom)
Next document	⌄ on the last page (top)	› on the last page, or >\	from any page, tooltip names the document (bottom)	› on the last page, or >\	from any page, tooltip names the document (bottom)
Score	2	4	4		

Conclusions:

Bottom placement wins — V2 and V3. The bar sits where the reader finishes the document, so moving on is the shortest path with no change of focus. Since 2026-10-23 neither shows reading progress, so both score 4.

V1 loses the most. Moving on is only possible from the toolbar at the top, and the vertical ⌄ / ^ don't read as "next document". For V1, consider duplicating "next" at the bottom (e.g. a floating › in the bottom-right corner — that is also a navigation panel, not document content).

V2 (new) matches V3's placement. Page and document buttons sit in the bottom pill where reading ends; like V3 it has no "how much is left" signal.

New (Should fix): on a document's last page the "next" control should signal that it leads to another document — e.g. a tooltip "Next document: Opening, quorum, and minutes.pdf" instead of "Next page", or an accent state on the button. This is a change in the panel itself, so it respects the constraint. **Status (V3): mostly fixed** — dedicated ‹‹ / ›› buttons with "Next document: <name>" tooltips sit in the bottom bar; the › button on the last page still says "Next page".

New (Should fix): the "N / 7" counter counts pages across the whole board book, while only one document is on screen. Show the local position too, e.g. "4 / 7 · page 1 of 2".

New (Should fix): Figma mockups. In the "Navigation" section, page 2 of another document appeared right under the cover. The mockups have been updated to show only the current document.

05 Must Fix

Must fix

V1, mobile: the toolbar breaks. At 375px the document field shrinks to zero and the page field overlaps it. Tools after "Download" are cut off at the screen edge. Cause: `.docSelectWrap { flex: 0 1 272px; min-width: 0 }` in `BoardBookPageNavViewer.module.css`, and the toolbar has no mobile variant. See `screenshots/review-v1-mobile-375.png`. *Fix: below 768 move the tools into a "⋯" menu, and give the document field a* `min-width` *(e.g. 120px) or move it to a second row.* **Status: fixed.** Below 768 the tools moved into "⋯", the document field has `min-width: 120px`, and the toolbar uses tighter spacing. See `screenshots/fixed-v1-mobile-375.png`.

Old V2 (removed 2026-10-22): no position indicator. There is no page number or "N / 7" anywhere. The user only sees the document name (Nielsen heuristic #1, "Visibility of system status"). See `screenshots/review-v2-desktop-1280.png`, `review-v2-tablet-768.png`. *Fix: add an "N / 7" counter between the arrows or in the zoom card.* **Status: fixed.** A floating "[N] / 7" counter with an input field was added at the bottom centre. See `screenshots/fixed-v2-desktop-1280.png`.

Old V2 (removed 2026-10-22): controls hidden until hover. The palette, zoom and arrows have `opacity: 0` until hover or focus (`.floating` in `BoardBookFilmstripViewer.module.css`). On desktop an open document looks as if it had no tools. See `screenshots/review-v2-desktop-1280.png`. *Fix: show the controls permanently (as V3 already does), or keep at least the arrows and the counter visible.* **Status: fixed.** Hover-hiding was removed; the controls are always visible.

V3, mobile: the bottom bar is cut off. At 375px "Fit page", "Highlight" and "Comment" are not visible. The bar has `max-width: calc(100% - 32px)` but doesn't wrap or collapse its contents. See `screenshots/review-v3-mobile-375.png`. *Fix: below 768 keep ‹ N / 7 › and zoom, and move "Fit" and annotations into "⋯".* **Status: fixed.** Below 768 "Fit width / Fit page / Highlight / Comment" moved into a "⋯" menu that opens upward. For this, `ActionMenu` got an optional `placement="top"` prop. See `screenshots/fixed-v3-mobile-375.png`.

06 Should Fix

Should fix

V3: progress shows 100% until the pages have loaded. Status: fixed together with the behaviour change. Progress is now computed across the whole board book (documents before the current one + scroll within it) and recalculated via `ResizeObserver` whenever the content height changes. Original description: `updateProgress` only ran on scroll and on `sheetWidth` / `focus` changes. If page images load late, `scrollHeight == clientHeight` at start and the bar stays full until the first scroll. Reproduced in headless Chrome. See `screenshots/review-v3-desktop-1280.png`, `review-v3-tablet-768.png`. *Fix: recompute progress with a `ResizeObserver` on the page stack in `BoardBookReadingViewer.tsx`.*

Inconsistent "next page" semantics. V1 uses ⌄ / ⌃ (vertical), V2/V3 use ‹ ›, and the hint at the bottom shows ← → everywhere. *Fix: pick one pattern for all versions; for vertical scrolling ⌃ / ⌄ are logical, but then the hint should be ↑ ↓.*

V1, mobile: the sheet doesn't fit. The sheet width is a fixed 635px (`calc(635px * zoom)`), so 375px needs horizontal scrolling. V2/V3 already use `min(635px, 100%)`. See `screenshots/review-v1-mobile-375.png`. *Fix: apply the same `min()` in V1.*

The zoom percentage means different things. V3 on mobile shows "54%" by default (relative to 635px). *Fix: compute the percentage relative to the fitted sheet, or show "Fit" when the sheet is fitted.* **Status (V2): fixed** — the new V2 shows "Fit" when fitted.

V3: the "Focus mode" icon is ambiguous. The DS `expand` icon (⌃⌄) reads as "expand / collapse". *Fix: use `arrow-resize-diagonal` as in V1/V2, or an icon such as `eye-opened`.* **Status: fixed.** Focus mode now uses `arrow-resize-diagonal` and sits in the bottom bar.

Different tool sets. V3 has no "Draw", V2 had no page field. This makes the comparison of versions not quite fair. *Fix: align the set of actions before the usability test.*

Hardcoded px in inline styles. `635px`, `960px`, `56px` and the 72px gutters are set in TSX (`style={{ width: … }}`), so stylelint can't see them. This breaks the CLAUDE.md "tokens only" rule. *Fix: move base sizes into module CSS variables or tokens.*

Code duplication. `useMediaQuery`, `FORMAT_ICON` and the `.docSelect*` styles are copied into 3 viewers (the old V2 viewer was deleted on 2026-10-22). *Fix: once a direction is chosen, keep one viewer and move shared parts into `PageStack`/shared.*

07 Version 2 (new) — re-review 2026-10-22

See `screenshots/review-v2-desktop-1280.png`, `review-v2-tablet-768.png`, `review-v2-mobile-375.png`, `review-v2-mobile-sheet-375.png`. No Must-fix items.

Should fix

Tablet portrait (768–1023) still shows both floating cards. They sit in the 16px gutters beside the sheet and don't cover text, but the phone's labelled "Document tools" sheet would be calmer there. *Fix: use the sheet below 1024px, not only below 768px.*

The phone trigger is icon-only (6 dots). Low-proficiency readers hesitate to open unlabeled menus (study I-08). *Fix: add a text label ("Tools") or use a labelled DS secondary button.*

No reading progress (V2 and V3). The bar shows "N / 7" only, so the end of a document is signalled by the page number alone. *Fix: "page 1 of 2" next to the counter (study R3).*

Could improve

Pill buttons are 28px (size S). Below the 44px touch recommendation on phones. *Suggestion: size M in the pill below 768px.*

Three control zones on desktop. The view card and the palette could merge into one card, or move into the top bar, to approach V3's single bar.

What works well

One bottom pill for page and document stepping, with distinct icons (‹ › pages, |< >| documents) and named tooltips.

The agenda open beside the document by default — the fastest way to jump to an agenda item.

The phone "Document tools" sheet: DS bottom Drawer, every row labelled, selected state shown, the sheet closes after a choice.

Consistent pressed state (hover fill) across the palette, cards and sheet; "Fit" replaces a misleading 100%.

08 Could Improve Could improve

Touch targets. 32×32 buttons (DS M) are below the recommended 44×44 on mobile. *Suggestion: size L, or a larger hit area below 768.*

Keyboard hint. It is `aria-hidden`, so screen readers don't announce it, and "← →" means nothing to trackpad and touch users. *Suggestion: hide it on touch devices; for keyboard users add a "Keyboard shortcuts" item to a menu.*

Dark mode. The test page has no theme toggle, so it couldn't be checked. All colours use tokens, so the risk is low. *Suggestion: add a theme toggle to the test page header.*

Tabs at 375px scroll horizontally and "Reports" is cut off. Acceptable for a test page.

09 What Works Well Keep

Continuous scrolling within a document (`PageStack`): a document's pages follow each other, the position is derived from scrolling, and navigation scrolls to the target. The reader moves to the next document only through the navigation panel ("next page" on the last page, the document field, the keyboard), as on the platform. This works the same in every version.

Real document and content. The Q4 pack with a matching agenda makes the comparison realistic.

The document field is the same in every version and opens a menu with agenda items and documents. Easy to find.

DS discipline in CSS. Colours, spacing and shadows come from tokens (`--elevation-100` on floating surfaces), `npm run verify` is clean for `src/test-versions`, buttons have `aria-label` and tooltips.

V3: one persistent control zone and a quiet top bar — the best balance of "document in focus / controls at hand".